Rahul Rathod

AI ENGINEER

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SUMMARY

AI Engineer in the trading department at Flutter UK & Ireland, one of the world's largest online betting and gaming operators. I build production AI systems for traders, including end-to-end Request a Bet automation (LLM request parsing, knowledge-graph prechecks, odds generation and compliant response generation), odds boost automation, and an experimentation platform with data-quality evals and RAG over experiment data. Founded and lead an internal AI Accelerator that embeds AI into how traders, analysts and risk specialists work. 16+ years of software engineering delivering reliable systems at scale in fast-paced, regulated environments.

CORE SKILLS

Languages TypeScript (Node.js), Python, SQL

Agents & LLMs Agent harnesses, tool use and multi-agent orchestration, Model Context Protocol (MCP) servers and gateways, Claude Agent SDK, OpenAI Agents SDK, LangGraph, PydanticAI; context engineering, prompt caching, structured output; Anthropic Claude, OpenAI GPT, Google Gemini, open-weight models (Llama, Qwen, DeepSeek); fine-tuning (LoRA)

Recommendation, Search & Ranking Real-time personalisation, two-stage retrieval and learned ranking (LightGBM LambdaRank), hybrid search (BM25 + vector), knowledge graphs, embeddings, RAG over structured and unstructured data, vector databases (pgvector, OpenSearch, FAISS)

Evaluation & Experimentation Offline metrics (NDCG, recall, coverage), A/B and champion/challenger testing, versioned eval sets with regression gates, data-quality and sanity evals, LLM-as-judge, groundedness checks, guardrails, tracing (Langfuse, LangSmith)

Cloud & Platform AWS (Lambda, ECS/Fargate, API Gateway, DynamoDB, S3, Bedrock, SageMaker, CloudWatch, X-Ray), infrastructure as code (AWS CDK, Terraform), Docker, Kafka, Redis, FastAPI, PostgreSQL, CI/CD (GitHub Actions, Jenkins), OpenTelemetry, Grafana; latency budgets and load testing

Leadership AI transformation and enablement, technical leadership and mentoring, cross-functional delivery, stakeholder management

EXPERIENCE

AI Engineer – Automation, Transformation and AI **Jun 2026 – Present**

Software Engineer **Jan 2023 – Jun 2026**

Flutter UK & Ireland (UKI Trading) *Leeds, UK*

- **Request a Bet automation.** Built the end-to-end pipeline that turns a customer's free-text bet request into a priced, ready-to-send offer. An LLM parses the request into structured selections, prechecks validate each leg against a knowledge graph of events, markets and participants, odds are generated for the combination, and the reply is written in the brand's voice within safer gambling and compliance standards.
- **Odds boost automation.** Automated the boost workflow end to end with AI built into each step, replacing manual, repetitive trader work with a consistent, auditable process.
- **Experimentation at scale.** Working with data science to redesign how experiments are run: a new data flow that lets experiments scale, eval suites that check data sanity before any result is trusted, and a RAG layer over experiment data so stakeholders can ask questions in plain English and get grounded answers.
- Founded and lead the department's AI Accelerator, a programme that embeds AI into day-to-day work for all traders and risk analysts and drives measurable, department-wide AI adoption.
- Own the full lifecycle of each AI system, from problem discovery and prototyping through evaluation, release, monitoring and iteration, with guardrails suited to a high-stakes, regulated domain.
- Partner with traders, product, risk and compliance to prioritise AI initiatives and ship tooling that is reliable, observable and cost-aware rather than one-off proofs of concept.

Assistant Consultant (via TCS) **Jan 2020 – Jan 2023**
Marks & Spencer *London, UK*

- Led delivery for Click & Collect and loyalty features across UK & ROI; integrated account, payments, and promotion APIs.
- Diagnosed and resolved a critical login regression impacting thousands of daily users; restored access and reduced support tickets.
- Worked closely with backend teams to consume Kafka-driven personalization feeds; partnered with analytics to enable data-informed product decisions.

Assistant Consultant (via TCS) **May 2017 – Dec 2019**
Best Buy *Chennai, India*

- Delivered modules for product search, recommendations, and dynamic pricing; integrated with Java microservices and caching layers.
- Enhanced responsiveness by optimizing API usage and background tasks; improved stability via monitoring and targeted fixes.

Technical Lead **May 2012 – Jul 2016**
Capgemini *Mumbai, India*

- Delivered enterprise-grade mobile apps for clients including 3M, ING, and Dako, achieving 4.5+ app store ratings on average.
- Managed distributed teams and QA/release pipelines, reducing post-release issues by 30% through performance and stability work.

Software Engineer **Apr 2011 – Jun 2012**
Persource *Ahmedabad, India*

- Built 15+ apps for global clients including Yahoo and Sundance Film Festival; owned full-cycle delivery from design to release.

Software Engineer **Sep 2010 – Apr 2011**
BYPT *Ahmedabad, India*

- Developed live sports scoring apps with real-time updates and performant UI, recognized for usability and stability.

SELECTED AI PROJECTS

Independent, shipped projects — full write-ups at raaulc.com/projects.

Real-time Game Recommender — Two-stage recommender (item-kNN and SVD retrieval, LightGBM LambdaRank ranker) taken from a Python research prototype to a TypeScript service. Ranks 23% better (NDCG@10) than the best single model on 3.15M plays from 53k players, at 3.87 ms p99. Updates from play events in real time, applies cooldown, exclusion and diversity guardrails, explains every pick, and is tested for parity with the prototype. Live demo at raaulc.com/projects/realtime-recs. *TypeScript, Node, Python, LightGBM, SciPy*

AI Paper Radar — Daily ranking pipeline over new arXiv AI papers: scores every paper for relevance and novelty with a typed-decision model, ranks them, and publishes a browsable archive. Runs unattended on a scheduled GitHub Actions workflow with retries and per-category fault tolerance. *Python, Jev, GitHub Actions, arXiv API*

Self-RAG Pipeline — Stateful LangGraph workflow where the LLM grades its own retrieved documents, checks for hallucinations, and validates its answer before responding. *LangGraph, LangChain, Chroma, Groq, HuggingFace*

EDUCATION

Bachelor of Engineering, Electronics & Communication May 2009
Dharmsinh Desai University — Gujarat, India